Wave Propagation and Energy

Local conservation, redshift as a change of bookkeeping, and the disappearance of the dark sector in the static T^4 cosmos

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Abstract

A propagating wave *always* carries a well-defined, positive energy: electromagnetically through the Poynting flux, quantized as $E = \hbar\omega$ per quantum, mechanically through radiation pressure. This document cleanly separates two levels that are constantly conflated in popular accounts: the *local* energy balance (always satisfied, $\partial_t u + \nabla\cdot \mathbf{S} = 0$ resp. $\nabla_\mu T^{\mu\nu} = 0$) and the *global* balance, which in curved, time-non-stationary spacetime is not uniquely definable. The widespread fallacy "energy is not conserved" (meaning: *destroyed*) mistakes a level of bookkeeping for physical destruction. In the FRW model the non-definability of total energy follows from the absence of a timelike Killing field (Noether); it is not a violation of local physics. In the FFGFT reading $\hbar$ and c are SI conversion factors; the physical content lies in the dimensionless ratios ($\tilde{T}\cdot m = 1$). The measured absolute energy is observer-dependent, the ratio structure invariant. On a *static* compact 4-torus T^4 the time-translation symmetry exists again; by Noether the total energy is then well-defined and conserved. Redshift thereby becomes a pure scale- and observer-rebooking rather than metric stretching of space, and the main motivation for dark energy (accelerated expansion) as well as part of the motivation for dark matter (CMB peak fit) disappears – the relevant structure follows in FFGFT from the T^4 geometry and from ξ (Doc. 267/268). This cosmological conclusion is framed at the status of the degeneracy thesis (Doc. 267): on the *data* both models are equivalent, on the theoretical *structure* FFGFT is the more economical reading (no dark sector, global energy conservation). R_H is here a shared empirical scale reference, not a free parameter – not an empirical knock-out of ΛCDM, but an economy argument.

.1 Framing: why this separation is necessary

The statement "a wave without energy would not be a wave" is trivially true and yet the starting point of a stubborn misunderstanding. In the cosmological discussion the correct technical observation "in an expanding universe there is no globally unique total energy" is regularly turned into the false conclusion "energy is destroyed". The two live on different levels.

Definition .1.1 (Two levels of balance). **Local balance** The continuity equation of the energy–momentum tensor, $\nabla_\mu T^{\mu\nu} = 0$, resp. in flat space $\partial_t u + \nabla\cdot \mathbf{S} = 0$. A statement about *flux*: energy vanishes nowhere, it flows. Always satisfied.

Global balance The existence of a unique, conserved *total energy* of the system. By Noether tied to time-translation symmetry; in non-stationary curved spacetimes not uniquely definable.

Warning .1.2 (The category error). "Globally not uniquely definable" is a statement about the *bookkeeping level*. "Physically destroyed" would be a violation of local physics – and that never occurs. Equating the two is the actual error.

This document works out the short note of the same title, substantiates the individual steps, and embeds them in the corpus: in the units question (Doc. 166, 182, 230), in the reduction structure (Doc. 270), and in the cosmological degeneracy (Doc. 267/268).

.2 Local energy balance – always satisfied

Electromagnetic field

For the electromagnetic field the energy density and the Poynting vector are [2, 3]

$$u = \frac{\varepsilon_0}{2}\,|\mathbf{E}|^2 + \frac{1}{2\mu_0}\,|\mathbf{B}|^2, \qquad \mathbf{S} = \frac{1}{\mu_0}\,\mathbf{E}\times\mathbf{B}, \tag{1}$$

with $\mathbf{S}$ as the energy flux density. The energy continuity equation reads

$$\frac{\partial u}{\partial t} + \nabla\cdot\mathbf{S} = -\,\mathbf{J}\cdot\mathbf{E}, \qquad \text{in vacuum} \quad \frac{\partial u}{\partial t} + \nabla\cdot\mathbf{S} = 0. \tag{2}$$

It is a *local* conservation statement: any decrease of the energy density at a point is exactly balanced by outflowing flux. Energy vanishes nowhere.

Quantization and radiation pressure

When quantized, each quantum carries

$$E = \hbar\omega, \qquad p = \hbar k = \frac{E}{c}, \qquad P_{\text{rad}} = \frac{S}{c}, \tag{3}$$

with momentum p and radiation pressure P_{rad}. As long as the wave propagates, energy flows – with measurable flux density, measurable pressure, measurable photoelectric effect.

Insight .2.1 (Operational content)**.** The local balance is non-negotiable because it is coupled to three independent measurable quantities: flux density S, radiation pressure $P_{\text{rad}} = S/c$, and the photoelectric threshold via $E = \hbar\omega$. A "propagating wave without energy" would contradict all three.

.3 Redshift: observer-dependent, not a loss into nothing

Under cosmological redshift the measured wavelength scales; a given observer measures $E = hc/\lambda$ correspondingly smaller. This does *not* mean that energy disappears, but that the question "how much energy" is frame- resp. observer-dependent.

Proposition .3.1 (Frame dependence of the photon energy)**.** *The photon energy is the time component of a four-momentum, $E = p^\mu u_\mu$, i.e. the scalar product with the observer's four-velocity u_μ. It is therefore not an invariant quantity but observer-dependent. An observer comoving at the time of emission measures the full energy; an observer in relative motion measures a different value. What remains* invariant *are ratios (frequency, energy, mass ratios).*

Energy here is thus a relative, observer-dependent quantity – not a destroyed one. This statement is model-independent (it holds in ΛCDM just as in FFGFT); what differs between the models is the *interpretation* of z (Section .7).

.4 The global fallacy – Noether precisely

Theorem .4.1 (Noether: energy conservation from time-translation symmetry)**.** *A globally conserved total energy exists if and only if the spacetime possesses a timelike Killing field ξ^μ, i.e. a time-translation symmetry $\nabla_{(\mu}\xi_{\nu)} = 0$ [1, 6, 5].*

An expanding FRW universe possesses no timelike Killing field, hence no global time-translation symmetry, hence no uniquely defined global total energy. This is a statement about the *non-existence of a unique global balance*, not about physical destruction. Locally,

$$\nabla_\mu T^{\mu\nu} = 0 \tag{4}$$

always holds.

Remark .4.2 (Where this usually goes wrong). The common crude error reads “energy is simply not conserved” – meaning destroyed. The correct statement is: in FRW the *global* bookkeeping is not unique because the symmetry argument of Theorem .4.1 does not apply. Equation (4) is unaffected by this.

.5 Absorption: energy is converted, not destroyed

Upon absorption the wave gives up its energy and changes form: photoelectric effect, heating, above threshold pair production. The kinematics requires a second partner: a single free photon cannot create a pair because of simultaneous energy and momentum conservation.

$$\gamma\gamma \to e^+e^- \quad \text{(Breit–Wheeler [4])}, \qquad \gamma + N \to e^+e^- + N \quad (E \geq 2m_ec^2). \tag{5}$$

Here too the local balance (2)/(4) remains satisfied at all times: the energy changes form, not existence.

.6 FFGFT view: energy as a partly unit-bookkeeping quantity

In FFGFT $\hbar$ and c are SI conversion factors, not ontological primitives; the physical content lies in the dimensionless ratios. The fundamental relation

$$\tilde{T} \cdot m = 1 \qquad \text{(natural units } \hbar = c = 1) \tag{6}$$

makes this explicit: $E \leftrightarrow 1/\tilde{T}$ and $m \leftrightarrow 1/\tilde{T}$.

Insight .6.1 (The ratio structure is invariant)**.** “How much energy” depends on the unit and observer convention; the ratio structure (mass, frequency, energy ratios) is invariant. This is exactly consonant with Proposition .3.1: the absolute value of the measured energy is observer-dependent, the invariant ratios remain.

In this reading “light → matter” is the binding of free phase (light) into mode content (mass): the energy changes the level of accounting, not its existence. The representational side of this statement – that the same 4D content is losslessly translatable between T^4 geometry and Hilbert-space language – is worked out in Doc. 230; the reduction chain $T^4 \to T^0$ with the duality decompactification of the mass dimension via $\tilde{T} \cdot m = 1$ in Doc. 270.

Remark .6.2 (Unit bookkeeping instead of ontology). The label “partly unit-bookkeeping” is to be read precisely: the *flux* and the *pressure* (Section 3) are operational, unit-independent facts. What hangs on the convention is solely the assignment of an absolute value “energy” to a given mode content – the choice of scale in which one counts.

.7 Cosmological consequence: static T^4, no dark sector

FFGFT lives on a compact, *static* 4-torus T^4. There is no metric expansion of space – and with it the motivation for dark energy (Λ) disappears, as does part of the motivation for dark matter. This has an immediate, simplifying effect on the energy discussion above.

Proposition .7.1 (Global energy conservation is restored). *A static T^4 background possesses a timelike time-translation symmetry. By Theorem .4.1 the total energy is then well-defined and conserved. The fallacy in the context of Theorem .4.1 ("no unique global energy") is a consequence of the* expansion assumption*; without it one need not retreat to "globally undefinable" in the first place.*

1. Redshift = change of bookkeeping, not stretching of space. Without expansion, redshift is not a stretching of the wavelength by growing space. It is the scale- and observer-rebooking encoded in $\tilde{T} \cdot m = 1$ (6): the energy and time scale of the distant source is referenced differently relative to the local standard. The photon carries its energy throughout (Sections 3); z is the conversion between source and observer bookkeeping – exactly the observer-dependent, relative energy of Proposition .3.1, only without the FRW apparatus.

2. No dark sector needed. ΛCDM needs dark energy to drive an accelerated expansion, and dark matter inter alia to fit the CMB peak structure. If expansion is removed, the main task of dark energy is gone. The relevant structure – e.g. the position of the CMB peaks – follows in FFGFT from the T^4 geometry and from ξ instead of from an expansion-plus-dark fit: the peak ratios are derived in Doc. 268 from the degeneracies $g(|\vec{n}|^2)$ of the T^4 modes, the scale ladder (Casimir/CMB/H_0) in Doc. 166.

Table 1: Theoretical cost of the two readings of the *same* data.

	ΛCDM pays	FFGFT
Accelerated expansion	dark energy Λ required	not needed (no expanding space)
CMB peak fit	dark matter, inter alia	from $T^4 + \xi$ (Doc. 268)
Global energy	not conserved (FRW, no Killing field)	conserved (Noether, static T^4)
"energy disappears"	lives here as a fallacy	dissolves

Warning .7.2 (Status: degenerate on the data, asymmetric in cost). On the *data* level the models are degenerate (Doc. 267): SNe Ia, BAO, and CMB temperature do not distinguish between metric expansion (ΛCDM) and fractal time unfolding (FFGFT) – the same data, two readings. On the *structure* level, however, the costs are *not* symmetric (Tab. 1): ΛCDM carries dark energy and dark matter as free components and gives up global energy conservation; FFGFT pays none of these. Under empirical equivalence, theoretical economy decides – and there FFGFT is the more economical reading. This is a parsimony argument (Occam), not an empirical knock-out of ΛCDM.

Remark .7.3 (R_H is a scale reference, not an input). R_H *need not* follow from ξ – it is a scale reference like any system-specific upper bound $R_{\text{Torus}}(m) = \hbar/mc$ (Doc. 182). The relation Planck length↔ R_H (exponent $41/4$, $R_H/L_0 = 6.49 \times 10^{64}$) is derived from first principles by *no* framework: why the universe is $\sim 10^{60}$ Planck lengths across is predicted by no theory. ΛCDM fixes the relation via the measured H_0 – itself circular in the sense of the degeneracy (Doc. 267); FFGFT adopts the established value for comparability and writes it in ξ notation (Doc. 190 P35/P36). This is a *shared empirical reference point* of both models, not a free parameter and not an FFGFT-specific gap. In FFGFT, moreover, $\hbar, c$ are SI conversion and G

is derived ($\xi \to \{m_e, \ldots\} \to G \to \ell_P$); a "measured" dimensional anchor in the standard-physics sense does not even exist here. The only asymmetry remains the dark sector, carried by ΛCDM alone.

.8 Methodological placement

The "local vs. global" separation introduced in Definition .1.1 is structurally the same pattern as elsewhere in the corpus: a deterministic, locally fully determined structure whose global or absolute accounting depends on the chosen level of resolution or convention (cf. Markov determinism, Doc. 038). The error "energy is destroyed" arises when one reads a missing global bookkeeping as a physical statement – exactly the kind of fallacy the correction register (Doc. 190) is written to guard against.

Insight .8.1 (What this document shows)**.** (i) A propagating wave always carries positive energy; (ii) locally, energy is always conserved; (iii) redshift is observer-dependent, not a loss; (iv) "globally non-unique" $\neq$ destroyed; (v) in a static T^4 the total energy is even globally conserved and z a pure rebooking. On the *data*, ΛCDM and FFGFT are degenerate; on the *structure*, FFGFT is the more economical reading (Tab. 1). The cosmic scale R_H is a shared empirical reference point, not a free parameter – the asymmetry lies solely in the dark sector.

.9 Summary

- Propagating wave $\Rightarrow$ always positive energy (Poynting (1), $E = \hbar\omega$ (3)).
- Locally, energy is always conserved ($\partial_t u + \nabla \cdot \mathbf{S} = 0$ (2); $\nabla_\mu T^{\mu\nu} = 0$ (4)).
- Redshift is observer-dependent (Proposition .3.1) – not a loss into nothing.
- No unique global energy (no time symmetry, Theorem .4.1) $\neq$ energy destroyed.
- FFGFT: energy is partly unit bookkeeping; physically it is the ratios ($\tilde{T} \cdot m = 1$ (6); Doc. 230/270).
- Static T^4 cosmos (no expansion): global energy conservation holds (Proposition .7.1), redshift is pure rebooking, the dark sector is not needed. Degenerate on the data (Doc. 267/268/166), more economical in cost (Tab. 1); R_H is a shared scale reference, not a free parameter – the asymmetry is solely the dark sector.

Connections. Doc. 270 (projection chain $T^4 \to T^0$, duality decompactification via $\tilde{T} \cdot m = 1$), Doc. 230 (lossless translation $T^4 \leftrightarrow$ Hilbert space), Doc. 267 (cosmological degeneracy ΛCDM vs. FFGFT), Doc. 268 (CMB peaks from T^4), Doc. 166 (Casimir/CMB/H_0 as a ξ power ladder), Doc. 182 (maximal scale R_H, calibrated), Doc. 190 (correction register, P35/P36).

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